



Diabetes & Metabolic Syndrome: Clinical Research & Reviews

journal homepage: www.elsevier.com/locate/dsx

The effects of low carbohydrate and high protein diet on the anthropometric indices, blood pressure, metabolic factors, and hormones related to metabolism: A systematic review and meta-analysis

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ARTICLE INFO

Keywords:

High-protein
Low-carbohydrate
Anthropometric
Glycemic
Lipid

ABSTRACT

Background and aims: The present study aimed to show the pooled effect of a low-carbohydrate high-protein diet (LCHPD) as a practical dietary approach on metabolic factors and anthropometric variables.

Methods and materials: A systematic search based on related keywords was performed from electronic databases, including PubMed, Scopus, Web of Science, and Google Scholar from inception until November 2024. The quality of the articles was evaluated using the Cochrane tool. Statistical analysis was conducted using STATA software.

Results: Sixteen articles were entered into the quantitative analysis. In total, 2915 participants (treatment: 1490 and control: 1425) entered the statistical analysis. Based on the statistical analysis LCHPD had desirable effects on the weight (SMD = −0.27; 95 % CI: −0.35, −0.19), systolic blood pressure (SBP) (SMD = −0.23; 95 % CI: −0.31, −0.25) diastolic blood pressure (DBP) (SMD = −0.12; 95 % CI: −0.21, −0.04), fasting blood sugar (FBG) (SMD = −0.34; 95 % CI: −0.43, −0.24), triglyceride (TG) (SMD = −0.29; 95 % CI: −0.38, −0.19), leptin (SMD = −0.57; 95 % CI: −0.68, −0.46), and, CRP (SMD = −0.38; 95 % CI: −0.49, −0.27), while it impressively increase serum level of total cholesterol (SMD = 0.36; 95 % CI: 0.26, 0.45), adiponectin (SMD = 0.40; 95 % CI: 0.29, 0.52), resistin, and (SMD = 1.08; 95 % CI: 0.96, 1.20). In addition, subgroup analysis was conducted based on the age of participants, duration of treatment, and type of disease.

Conclusion: Overall findings of the present study exhibited that LCHPD had a significant effect on the anthropometric indices, blood pressure, and metabolic factors.

1. Introduction

The global increase in obesity and metabolic disorders has posed a major public health challenge for decades [1]. Recent studies suggest that almost one in every four individuals is affected by metabolic syndrome (MS) [2]. MS is characterized by a cluster of metabolic abnormalities, including obesity, hyperglycemia, hyperlipidemia, and hypertension, which substantially elevate the risk of increased risk of non-communicable diseases particularly type 2 Diabetes (T2D) and cardiovascular disease [3]. This condition imposes a considerable

economic burden, with the global cost of T2DM projected to reach \$622.3 billion annually by 2030 [4]. Genetic predisposition, accumulation and improper fat distribution, insulin resistance, systemic inflammation and environmental factors (e.g., insufficient physical activity and improper dietary patterns) are considered to be the primary causes of MS [5–8]. Moreover, numerous studies have confirmed that MS is closely associated with various anthropometric measures, especially central obesity indices such as waist circumference and waist-to-hip ratio. These indices are useful tools in both clinical and research settings and have shown strong predictive value for metabolic

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<https://doi.org/10.1016/j.dsx.2025.103352>

Received 10 February 2025; Received in revised form 3 December 2025; Accepted 5 December 2025

Available online 6 December 2025

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abnormalities, including insulin resistance, hypertension, and dyslipidemia [9–11]. Diagnostic approaches are different but majorly focused on abdominal obesity, hyperglycemia, dislipidemia, and hypertension. Medications used to treat MS can help ease some of its symptoms, but do not address its underlying causes. These drugs, such as anti-hypertensive agents, antihyperglycemic agents, and statins, typically focus on lowering blood pressure, blood sugar, and blood fats, while key factors like obesity and insulin resistance remain largely uncontrolled [12]. Moreover, common side effects, including electrolyte imbalance, hypoglycemia, gastrointestinal issues, myalgia, and fatigue, can reduce patient adherence to medication therapy [13–15]. For this reason, adjusting dietary habits, physical activity and reducing excess visceral fat are important. Focusing on nutrition are considered as a fundamental intervention for managing MS.

One of the emerging dietary patterns under investigation is the low-carb, high-protein diet (LCHPD), characterized by an intake of about 10–30 % of daily calories from carbohydrates and about 30–40 % from protein [16,17]. Previous studies have indicated that LCHPD enhanced fat oxidation, improves insulin sensitivity, stabilizes blood sugar, reduces inflammation, and regulates appetite through hormonal pathways [18–21]. Metabolic hormones, such as adiponectin, leptin, and resistin, can significantly impact physiological processes when affected by low-carbohydrate diets [22,23]. Leptin, a hormone produced by fat tissue, rises as fat stores grow [24]. It signals satiety to the hypothalamus, stimulating metabolic processes and thermogenesis, and even supporting immunity and reproduction systems [25–28]. Another hormone secreted by adipose tissue is adiponectin, which, unlike leptin, decreases as the volume of adipose tissue increases [29]. Adiponectin plays a crucial role in preventing metabolic diseases, including Type 2 Diabetes (T2D), cardiovascular diseases, and hepatic steatosis through its anti-inflammatory effects, its ability to increase insulin sensitivity, and its role in regulating lipid metabolism [30–33]. Resistin, another adipokine which secreted by macrophages within adipose tissue, contributes to insulin resistance and inflammation, making it a key predictor for the onset of metabolic diseases [34]. Despite its significant impact on metabolic health, there is still limited research on how different macronutrient patterns affect its levels and activity.

Previous meta-analyses have examined long-term weight maintenance after LCHPD [35], comparing low-carbohydrate diets against low-fat diets [36–40] and even interventions with either a high-protein diet [41] or a low-carbohydrate diet [42–45] on their own. However, none of these studies have specifically addressed the short-term effects of low-carbohydrate and low-fat diets. They focus on long-term endpoints or isolate the effect of one macronutrient, often neglecting the combined impact of carbohydrate restriction and increased protein intake in the short term. Therefore, to the best of our knowledge, this systematic review and meta-analysis is the first to assess clinical interventions involving a low-carbohydrate, high-protein diet on a broad range of metabolic and physiological outcomes.

This study aims to evaluate the impact of LCHPD on diverse metabolic parameters, including anthropometric parameters, glycemic control indicators, lipid profile, blood pressure, inflammation, and metabolism-related hormones. By addressing an underexplored area in nutritional research, this review will provide evidence to guide short-term dietary interventions in both clinical and public health settings.

2. Methods

The current study was conducted based on the guidelines provided in the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) [46], and also followed the instructions outlined in the Cochrane Handbook for Systematic Reviews of Interventions [47]. The current research was registered in the International Prospective Register of Systematic Reviews (PROSPERO), and the obtained number is CRD42025633417.

2.1. Search strategy

A comprehensive literature search was carried out in electronic databases including PubMed (<http://www.ncbi.nlm.nih.gov/pubmed>), Scopus (<http://www.scopus.com>), and Web of Science (<http://www.webofscience.com>) from inception until November 2024 without any limitations, to obtain the relevant randomized clinical trials (RCTs) that investigated the effects of low carbohydrate and high protein diet on the anthropometric, metabolic, and hormonal variables in human samples. The search terms included combinations of keywords and MeSH terms related to Diet, High-Protein Low-Carbohydrate OR Atkins diet OR Modified Atkins diet AND Clinical Trials OR Cross-Over Studies OR Double-Blind Method OR Single-Blind Method OR Random Allocation OR RCT OR Intervention Studies OR intervention OR Controlled Clinical Trial OR randomized OR randomised OR random OR randomly OR Placebos OR assignment. In addition, a hand search was conducted on Google Scholar (<https://scholar.google.com/>), and a reference list of related articles was reviewed to ensure search results.

2.2. Selection study

All of the defined studies were entered into an EndNote file, and duplicate studies (based on the names of authors and journal) were deleted. Then, two independent reviewers (A.H.S and M.J.P) screened the title, abstract, and full text of the remaining articles. Any disagreement between the two researchers was unraveled by a third researcher (O.N). Clinical trial studies that determined the effects of low-carbohydrate and high-protein diets on the anthropometric variables, glycemic control, lipid profile, inflammation indices, and hormones in adults were included in the systematic review and meta-analysis. Exclusion criteria in the present study include observational studies (case-control, cross-sectional, cohort), quasi-experimental studies, articles that do not report sufficient data, research conducted on children, and non-English articles.

2.3. Data extraction

Full text of the included studies in the analysis was screened and important relevant data including the author's name, year of publication, country, type of intervention, mean age of participants, number of subjects, medical condition of participants, duration of treatment, mean and standard deviation (SD) of the outcome, and design of study such as blinding and cross-over.

2.4. Quality assessment

The quality of the methodology and findings of entered studies were evaluated by two independent researchers (O.N) using the Cochrane tool for risk of bias assessment guidelines [47], this tool assesses the quality of research according to the following items: random sequence generation, allocation concealment, blinding of participants & personnel, blinding of outcome assessment, incomplete outcome data, selective outcome reporting, and other sources of bias.

2.5. Data synthesis

All of the statistical analyses were performed by Stata software (StataCorp, College Station, TX, USA) version 15. A random-effects model (DerSimonian and Laird) was conducted to determine pooled effect sizes with 95 % confidence intervals (95 %CI) for each variable. The degree of heterogeneity was identified based on the I^2 statistic and associated p-value; I^2 more than 50 % and p-value less than 0.05 were considered significant. Subgroup analysis was performed based on the mean age of participants and duration of follow-up to investigate the effects of these factors on the pooled effect size. Sensitivity analysis was carried out to exhibit changes in the pooled effect size by the elimination

of each study. In addition, publication bias was assessed using Begg's and Egger's tests, as well as a funnel plot for visual examination of asymmetry.

3. Results

3.1. Study selection

The process of the selection study is presented in Fig. 1. At first, 1418 studies were obtained through searching in electronic databases, while 408 articles were duplicated and removed. Also, 967 studies were excluded due to irrelevant studies ($N = 877$), animal research ($N = 36$), not accessible ($N = 6$), and review articles ($N = 48$). Full-text of 43 articles reviewed; however, 27 studies were removed because of insufficient data ($N = 10$) and unrelated information ($N = 17$). So 16 articles were entered into the analysis. General characteristics of the entered studies are exhibited in Table 1. In total, 2915 subjects (treatment: 1490 and control: 1425) were included in the final analysis. The included studies were performed on obese participants [48–56], Type 2 Diabetes (T2D) [57,58], polycystic ovary syndrome (PCOS) [59,60], epilepsy [61], patients who underwent gastric bypass [62], and spinal cord injury (SCI) patients [63]. Moreover, the duration of treatment was different between studies from 6 to 104 weeks. The results of the quality assessment of the included studies are demonstrated in Table 2.

3.2. The effects of LCHPD on anthropometric measures and body composition

Supplementary Figure 1 revealed the effects of LCHPD on the anthropometric indices and fat mass (see Table 3). Overall, thirteen studies indicated the effects of LCHPD on weight. Based on the analysis, LCHPD had an impressive effect on weight ($SMD = -0.27$; 95 % CI: $-0.35, -0.19$, $P < 0.0001$), however, there is a high heterogeneity between included studies ($I^2: 94.9\%$, $P < 0.001$). Eight articles reported the effects of LCHPD on body mass index (BMI) and based on the analysis indicated a significant pooled effect size ($SMD = -0.59$; 95 % CI: $-0.71, -0.47$, $P < 0.0001$), but there was a high heterogeneity among included studies in the analysis ($I^2: 98.5\%$, $P < 0.001$). Based on the findings of included studies in the analysis, LCHPD caused a meaningful reduction in waist-to-height ratio (WHtR) ($SMD = -0.35$; 95 % CI: $-0.48, -0.21$, $P < 0.0001$), though there was a remarkable heterogeneity among studies ($I^2: 99.2\%$, $P < 0.001$). In contrast, statistical

analysis showed that LCHPD significantly increased the waist circumference (WC) ($SMD = 0.27$; 95 % CI: $0.17, 0.37$, $P < 0.0001$), although heterogeneity was considerable ($I^2: 98.4\%$, $P < 0.001$). The effects of LCHPD on the fat mass were in line with the effect of WC; in fact, the analysis demonstrated an increment in fat mass ($SMD = 0.41$; 95 % CI: $0.29, 0.52$, $P < 0.0001$), with high heterogeneity ($I^2: 98.0\%$, $P < 0.001$).

According to the sensitivity analysis, none of the included articles had a significant effect on the pooled effect sizes of weight, BMI, WC, WHtR, and fat mass. Moreover, there was not any publication bias for weight (Begg's $P = 0.627$ and Egger's $P = 0.955$), BMI (Begg's $P = 0.448$ and Egger's $P = 0.186$), WC (Begg's $P = 0.845$ and Egger's $P = 0.629$), WHtR (Begg's $P = 0.128$ and Egger's $P = 0.399$), and fat mass (Begg's $P = 0.586$ and Egger's $P = 0.531$).

3.3. The effects of LCHPD on blood pressure

The effect of LCHPD on systolic blood pressure (SBP) and diastolic blood pressure (DBP) is presented in Supplementary Fig. 2. (see Table 3). Six randomized clinical trials reported the effect of LCHPD on SBP; the pooled effect size showed that LCHPD had a significant effect on SBP ($SMD = -0.23$; 95 % CI: $-0.31, -0.15$, $P < 0.0001$), although there was heterogeneity between included studies ($I^2: 62.7\%$, $P = 0.002$). Also, LCHPD had a considerable effect on DBP ($SMD = -0.12$; 95 % CI: $-0.21, -0.04$, $P = 0.006$), and a high heterogeneity was observed among studies ($I^2: 95.3\%$, $P < 0.001$).

Sensitivity analysis exhibited that the exclusion of each study did not have any meaningful effect on the pooled effect size of SBP and DBP. In addition, the analysis did not demonstrate any publication bias for SBP (Begg's $P = 0.251$ and Egger's $P = 0.719$) and DBP (Begg's $P = 0.567$ and Egger's $P = 0.614$).

3.4. The effects of LCHPD on glycemic control

Supplementary Fig. 3 reports the effect of LCHPD on the glycemic indices. Statistical analysis indicated that LCHPD had a desirable effect on the level of fasting blood sugar (FBS) ($SMD = -0.34$; 95 % CI: $-0.43, -0.24$, $P < 0.0001$), but it was heterogeneous ($I^2: 90.1\%$, $P < 0.001$). As well as, LCHPD had a reducing effect on serum insulin ($SMD = -0.78$; 95 % CI: $0.87, -0.68$, $P < 0.0001$), although there was high heterogeneity between studies ($I^2: 95.2\%$, $P < 0.001$) (see Table 3). Also, it was shown that the effect of LCHPD on Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) is remarkable ($SMD = -0.26$; 95 % CI: $0.37, -0.16$, $P < 0.0001$), while there was heterogeneity between included studies ($I^2: 98.2\%$, $P < 0.001$).

Based on the analysis pooled effect sizes of FBG, insulin, and HOMA-IR were not changed by the elimination of one study. Publication bias was not significant for FBG (Begg's $P = 0.289$ and Egger's $P = 0.733$), insulin (Begg's $P = 0.383$ and Egger's $P = 106$), and HOMA-IR (Begg's $P = 287$ and Egger's $P = 0.159$).

3.5. The effects of LCHPD on lipid profile

The effect of the LCHPD on the lipid profile is presented in Supplementary Fig. 4. (see Table 3) Seven articles reported the effects of LCHPD on the level of total cholesterol (TC) ($SMD = 0.36$; 95 % CI: $0.26, 0.45$, $P < 0.0001$), although with a high heterogeneity ($I^2: 93.7\%$, $P < 0.001$). In line, the meta-analysis showed that LCHPD increases low-density lipoprotein-cholesterol (LDL-C) ($SMD = 0.21$; 95 % CI: $0.12, 0.31$, $P < 0.0001$), with heterogeneity ($I^2: 97.5\%$, $P < 0.001$). In contrast, the LCHPD had a desirable effect on the triglyceride (TG) ($SMD = -0.29$; 95 % CI: $0.38, -0.19$, $P < 0.0001$), with a high heterogeneity ($I^2: 83.6\%$, $P < 0.001$). Also, the LCHPD makes a substantial reduction in high-density lipoprotein-cholesterol (HDL-C) levels ($SMD = -0.33$; 95 % CI: $0.41, -0.24$, $P < 0.0001$), with high heterogeneity ($I^2: 94.5\%$, $P < 0.001$).

The pooled effect sizes of the lipid profile were not affected by the

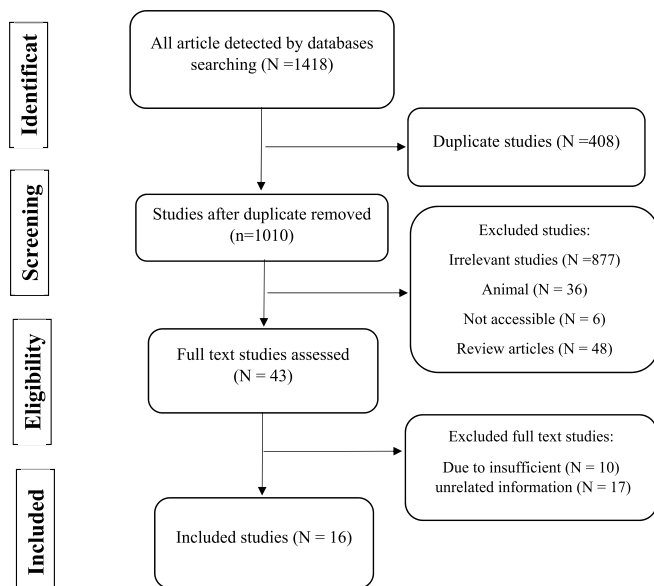


Fig. 1. Literature search and study selection process.

Table 1
Characteristics of trials included in the meta-analysis.

Studies	Country	Participant	Sample size		Intervention Duration (Weeks)	Age of Participants (Mean $\pm$ SD)		Intervention diet	Control diet
			Treatment	Control		Treatment	Control		
Richter et al. [57]	USA	T2DM	12	12	6	66.0 $\pm$ 7	62.0 $\pm$ 9	Isocaloric study: CRHPD: 30 %Carb, 30 %Pro and 40 %Fat	Isocaloric study: CD: 50 %Carb, 17 %Pro and 33 %Fat
Richter et al. [57]	USA	T2DM	34	32	6	66.0 $\pm$ 7	67.0 $\pm$ 9	Hypocaloric study: CRHPD: 30 %Carb, 30 %Pro and 40 %Fat	Hypocaloric study: CD: 50 %Carb, 17 %Pro and 33 %Fat
Li et al. [63]	USA	SCI	11	14	8	40.6 $\pm$ 12.6	44.2 $\pm$ 10.9	LCHPD: 40 %Carb, 30 %Pro and 30 %Fat	Regular Diet
Nadjarzadeh et al. [59]	Iran	PCOS	16	16	24	29.43 $\pm$ 6.60	28.87 $\pm$ 6.59	HHPPD: 40 %Carb, 30 %Pro and 30 %Fat	HSDP: 55 %Carb, 15 %Pro and 30 %Fat
Barry et al. [54]	USA	Overweight and Obesity	32	18	15	52.8 $\pm$ 2.1	48.4 $\pm$ 10.3	LCHFD: 5 %Carb, 30 %Pro and 65 %Fat	LFD: 63 %Carb, 13 %Pro and 23 %Fat
Primo et al. [52]	Spain	Obesity	136	132	36	GG:52.1 $\pm$ 6.3	GA:54.8 $\pm$ 3.1	HPLCLCD: 33 %Fat, 33 %Carb and 34 %Pro	SPLCD: 27 %Fat, 53 %Carb and 20 %Pro
De luis et al. [51]	Spain	Obesity	94	97	36	47.1 $\pm$ 0.1	47.1 $\pm$ 0.1	HPLCD: 40.3 %Fat, 40.7 %Carb and 27 %Pro	SHD: 39.1 %Fat, 42.3 %Carb and 19.6 %Pro
De luis et al. [51]	Spain	Obesity	43	36	36	47.1 $\pm$ 0.1	47.1 $\pm$ 0.1	HPLCD: 40.3 %Fat, 40.7 %Carb and 27 %Pro	SHD: 39.1 %Fat, 42.3 %Carb and 19.6 %Pro
Zare et al. [61]	Iran	Refractory Epilepsy	34	32	8	29.4 $\pm$ 8.8	27.2 $\pm$ 7.3	antiepileptic drugs + MAD: carbohydrates limited to 15 g/day, without any changes or limitations in calories and liquid in dietary pattern, it means 4–6 %Carb, 20–30 %Pro, and 60–70 %fat	antiepileptic drugs
De luis et al. [50]	Spain	Obesity	168	163	36	50.5 $\pm$ 13.1	49.9 $\pm$ 12.0	After 3 month: HPHD: 33 %Carb, 33 %Fat and 34 %Pro	SPHD: 53 %Carb, 27 %Fat, and 20 %Pro
De luis et al. [50]	Spain	Obesity	168	163	36	50.5 $\pm$ 13.1	49.9 $\pm$ 12.0	After 6 month: HPHD: 33 %Carb, 33 %Fat and 34 %Pro	SPHD: 53 %Carb, 27 %Fat, and 20 %Pro
De luis et al. [50]	Spain	Obesity	168	163	36	50.5 $\pm$ 13.1	49.9 $\pm$ 12.0	After 9 month: HPHD: 33 %Carb, 33 %Fat and 34 %Pro	SPHD: 53 %Carb, 27 %Fat, and 20 %Pro
Friedman et al. [49]	USA	Obesity	153	154	104	46.2 $\pm$ 9.2	44.9 $\pm$ 10.2	LCD: Participants were instructed to follow a diet limited carbohydrate intake but allowing unlimited fat and protein consumption. During the first 12 weeks of treatment, participants were instructed to limit carbohydrate intake to 20 g/d in the form of low-glycemic index vegetables. After the first 12 weeks, participants gradually increased carbohydrate intake (5 g/d per week) by consuming more vegetables, a limited amount of fruits, and eventually small quantities of whole grains and dairy products, until a stable and desired weight was achieved.	LFD with limited energy intake: 55 %Carb, 30 %Fat and 15 %Pro
Larsen et al. [58] 2011	Australia	Overweight/obese with type 2 diabetes mellitus	53	46	48	59.6 (57.5, 61.8)	58.8 (55.8, 61.7)	HPD: 30 %Pro and 40 %Carb	HCD: 15 %Pro and 55 %Carb
Foster et al. [56]	USA	Obesity and overweight	153	154	104	46.2 $\pm$ 9.2	44.9 $\pm$ 10.2	LCD: which limited carbohydrate intake but allowed unrestricted consumption of fat and protein. During the first 12 weeks of treatment, participants were instructed to limit carbohydrate intake to	LFD: 55 %Carb, 30 %Fat and 15 %Pro

(continued on next page)

Table 1 (continued)

Studies	Country	Participant	Sample size		Intervention Duration (Weeks)	Age of Participants (Mean $\pm$ SD)		Intervention diet	Control diet
			Treatment	Control		Treatment	Control		
								20 g/d in the form of low-glycemic index vegetables. After the first 12 weeks, participants gradually increased carbo-hydrate intake (5 g/d per week) by consuming more vege-tables, a limited amount of fruits, and eventually small quantities of whole grains and dairy products, until a stable and desired weight was achieved.	
Rolland et al. [55]	United Kingdom	Obesity	38	34	36	42.7 $\pm$ 13.1	39.9 $\pm$ 10.4	LCHPD: restricted to $\leq$ 40 g carbohydrate / day. The energy intake ranged from 800 to 1500 kcal, where an 800 kcal diet was composed of 20 % Carb, 40 %Pro, and 40 %Fat	LL: 36 %Carb, 36 % Pro, and 28 %Fat
Swenson et al. [62]	USA	Patient Underwent Gastric Bypass	19	13	48	41.7 $\pm$ 9.8	39.7 $\pm$ 7.6	HPLC diet: The South Beach Diet	LFD: patients were advised to consume 50–60 g of protein each day and limit fat intake to less than 35 g per day
Galletly et al. [60]	Australia	PCOS	12	13	24	33.0 $\pm$ 1.2	32.0 $\pm$ 1.2	HPLCD: 30 %Pro, 40 %Carb, 30 %fat	LPHCD: 15 %Pro, 55 %Carb, 30 %fat
De luis et al. [53]	Spain	Obesity	47	43	12	43.0 $\pm$ 14.3	42.1 $\pm$ 16.0	LCD: 38 %Carb, 26 %Pro, 36 %Fats	LFD: 52 %Carb, 20 % Pro, 27 %Fats
Foster et al. [48]	USA	Obesity	33	30	48	44.0 $\pm$ 9.4	44.2 $\pm$ 7.0	After 3 month: LCD: involves limiting carbohydrate intake with-out restricting consumption of fat and protein. For the first two weeks, carbohydrate intake is limited to 20 g per day and is then gradually increased until a stable and desired weight is achieved.	CD: 60 %Carb, 25 % Fat and 15 %Pro
Foster et al. [48]	USA	Obesity	33	30	48	44.0 $\pm$ 9.4	44.2 $\pm$ 7.0	After 6 month: LCD: involves limiting carbohydrate intake with-out restricting consumption of fat and protein. For the first two weeks, carbohydrate intake is limited to 20 g per day and is then gradually increased until a stable and desired weight is achieved.	CD: 60 %Carb, 25 % Fat and 15 %Pro
Foster et al. [48]	USA	Obesity	33	30	48	44.0 $\pm$ 9.4	44.2 $\pm$ 7.0	After 12 month: LCD: involves limiting carbohydrate intake with-out restricting consumption of fat and protein. For the first two weeks, carbohydrate intake is limited to 20 g per day and is then gradually increased until a stable and desired weight is achieved.	CD: 60 %Carb, 25 % Fat and 15 %Pro

PCOS: Polycystic Ovary Syndrome, SCI: Spinal Cord Injury, T2DM: type 2 diabetes mellitus, Carb: Carbohydrate, Pro: protein, CRHPD: Carbohydrate-reduced high-protein diet, CD: Conventional diet, LCHPD: Low carbohydrate high protein diet, HHPPD: Hypocaloric high-protein low-carbohydrate diet, HSDP: Hypocaloric standardize diet + placebo, LCHF: Low-carbohydrate high-fat diet, LFD: Low fat diet, HPLCLCD: High protein low carbohydrate low-calorie diet, SPLCD: Standard protein low-calorie diet, HPLCD: High-protein low-carbohydrate diet, SHD: Standard hypocaloric diet, HPHD: High protein hypocaloric diet, SPHD: Standard protein hypocaloric diet, LCD: Low carbohydrate diet, LFD: Low fat diet, HPD: High Protein diet, HCD: High Carbohydrate diet, LL: Lighter Life, MAD: Modified Atkins diet, HPLCD: High-protein/low carbohydrate diet, LPHCD: Low-protein high-carbohydrate diet.

Table 2
Quality assessment of studies included in the meta-analysis.

First Author	Random sequence generation	Allocation concealment	Blinding of participants & personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other sources of bias	Overall quality
Ritcher et al. [57]	L	L	H	H	L	L	L	High risk of bias
Li et al. [63]	L	L	L	L	L	L	L	Low risk of bias
Nadjarzadeh et al. [59]	L	L	L	H	L	L	L	High risk of bias
Barry et al. [54]	L	U	H	H	L	L	U	High risk of bias
Primo et al. [52]	L	L	H	H	L	L	L	High risk of bias
De luis et al. [51]	L	L	H	H	L	L	L	High risk of bias
Zare et al. [61]	L	L	H	H	L	L	L	High risk of bias
De luis et al. [50]	L	H	H	H	L	L	L	High risk of bias
Friedman et al. [49]	L	U	H	H	L	L	L	High risk of bias
Larsen et al. [58]	L	L	L	L	L	L	U	Low risk of bias
Foster et al. [56]	L	L	H	H	L	L	U	High risk of bias
Rolland et al. [55]	L	L	H	H	L	L	L	High risk of bias
Swenson et al. [62]	L	L	L	H	L	L	L	High risk of bias
Galletly et al. [60]	L	L	H	H	L	L	U	High risk of bias
De luis et al. [53]	L	U	H	H	L	L	U	High risk of bias
Foster et al. [48]	L	U	H	H	L	L	L	High risk of bias

Abbreviations: H, high risk of bias; L, low risk of bias; U, unclear risk of bias.

elimination of each study included in the analysis. Moreover, there was no significant publication bias for TC (Begg’s P = 0.451 and Egger’s P = 0.933), LDL-C (Begg’s P = 0.283 and Egger’s P = 583), HDL-C (Begg’s P = 0.483 and Egger’s P = 582), and, TG (Begg’s P = 0.327 and Egger’s P = 537).

3.6. The effects of LCHPD on hormones

Supplementary Fig. 5. Demonstrates the effect of LCHPD on the level of adiponectin, resistin, and leptin (see Table 3). Analysis revealed that the LCHPD significantly increased the level of adiponectin (SMD = 0.40; 95 % CI: 0.29, 0.51, P < 0.0001), but there is a high heterogeneity (I²: 97.7 %, P < 0.001). Also, the effects of the LCHPD on resistin were increasing (SMD = 1.08; 95 % CI: 0.96, 1.20, P < 0.0001), with a high heterogeneity (I²: 98.8 %, P < 0.001). While the LCHPD substantially decreased the level of leptin (SMD = −0.57; 95 % CI: 0.68, −0.56, P < 0.0001), however, there was a high heterogeneity among included studies (I²: 92.6 %, P < 0.001).

Sensitivity analysis on adiponectin, resistin, and leptin showed that excluding each of the entered studies did not change the pooled effect size. Moreover, Egger’s regression and Begg’s rank tests did not reveal statistically significant publication bias for adiponectin, resistin, and leptin (p > 0.05).

3.7. The effects of LCHPD on CRP

Statistical analysis indicated that LCHPD had a substantial effect on the C-reactive protein (CRP) level (SMD = −0.38; 95 % CI: 0.49, −0.27, P < 0.0001), but there was high heterogeneity between included studies (I²: 84.1 %, P < 0.001), **Supplementary Fig. 6.** (see Table 3). Sensitivity analysis revealed that the pooled effect size was not affected by any of the studies; also, there was no publication bias (Begg’s P = 0.384 and

Egger’s P = 307).

3.7.1. Subgroup analysis

Due to the high heterogeneity between included studies, we conducted a sub-group analysis based on the mean age of participants (<50 old and ≥50 old), duration of follow-up (<24 weeks and ≥24 weeks) and, type of disease (Obesity and other type of disease) to determine the source of heterogeneity. In terms of the age of patients, LCHPD was more effective in participants older than 50 years old in reducing weight, BMI, WHtR, SBP, FBS, insulin, HOMA-IR, and HDL-C, while it had a greater effect on the level of fat mass, DBP, TG, resistin, and, leptin in subjects younger than 50 years old. In regard to the duration of treatment, LCFPD had a bigger effect on weight, BMI, WC, fat mass, FBS, HOMA-IR, TC, LDL-C, and HDL-C in less than 24 weeks in comparison with more than 24 weeks, however, level of the insulin and WHtR reduced more in longer than 24 weeks of treatment. As well as, LCHPD in obese patients was more effective on variables such as BMI and insulin, while this dietary approach in other patients had a greater effect on weight, FBG, insulin, HOMA-IR, TC, and, LDL-C.

4. Discussion

This systematic review and meta-analysis is, to our knowledge, the first to comprehensively evaluate the effects of a LCHPD on metabolic, inflammatory, and hormonal markers. After a comprehensive analysis of the relevant literature, we concluded that LCHPD reduces weight, BMI, WHtR, SBP, DBP, FBG, serum insulin levels, HOMA-IR, TG, and CRP. Moreover, it decreases leptin and increases adiponectin, although it does not have a beneficial effect on other factors.

Our study results were further clarified through subgroup analysis, which effectively highlighted how the mean age of participants (<50 years and ≥50 years) and the duration of follow-up (<24 weeks and ≥24

Table 3

Pooled effect sizes of LCHPD on variables.

Variable	Group	No. of trials	SMD (95 %CI)	P-value	I ² (%)	P-heterogeneity
Weight	Total	20	-0.27 (-0.35, -0.19)	<0.0001	94.9	<0.0001
	Mean age					
	<50 old	12	-0.14 (-0.25, -0.02)	0.021	96.1	<0.0001
	≥50 old	8	-0.38 (-0.49, -0.27)	<0.0001	91.5	<0.0001
	Treatment duration					
	<24 weeks	6	-1.02 (-1.32, -0.73)	<0.0001	96.6	<0.0001
	≥24 weeks	14	-0.21 (-0.29, -0.13)	<0.0001	93.6	<0.0001
	Type of disease					
	Obesity	14	-0.19 (-0.27, -0.11)	<0.0001	94.1	<0.0001
	Other	6	-0.32 (-0.58, -0.05)	0.0200	91.1	<0.0001
BMI	Total	12	-0.59 (-0.71, -0.47)	<0.0001	98.5	<0.0001
	Mean age					
	<50 old	7	-0.51 (-0.71, -0.31)	<0.0001	97.2	<0.0001
	≥50 old	5	-0.64 (-0.79, -0.49)	<0.0001	99.2	<0.0001
	Treatment duration					
	<24 weeks	3	-2.59 (-3.08, -2.10)	<0.0001	98.3	<0.0001
	≥24 weeks	9	-0.46 (-0.58, -0.33)	<0.0001	98.5	<0.0001
	Type of disease					
	Obesity	10	-0.66 (-0.78, -0.54)	<0.0001	98.7	<0.0001
	Other	2	0.48 (-0.03, 0.93)	0.791	00.0	0.791
WC	Total	10	0.27(0.17, 0.37)	<0.0001	98.4	<0.0001
	Mean age					
	<50 old	5	0.10 (-0.09, 0.30)	0.313	97.1	<0.0001
	≥50 old	5	0.33 (0.21, 0.45)	<0.0001	99	<0.0001
	Treatment duration					
	<24 weeks	2	-0.52 (-0.88, -0.16)	0.005	46	0.173
	≥24 weeks	8	0.33 (0.23, 0.44)	<0.0001	98.7	<0.0001
	Total	5	-0.35 (-0.48, -0.21)	<0.0001	99.2	<0.0001
	Mean age					
	<50 old	2	-0.05 (-0.41, 0.30)	0.777	00.0	0.634
WHtR	≥50 old	3	-0.39 (-0.54, -0.25)	<0.0001	99.6	<0.0001
	Treatment duration					
	<24 weeks	2	-0.05 (-0.41, 0.30)	0.777	00.0	0.634
	≥24 weeks	3	-0.39 (-0.54, -0.25)	<0.0001	99.6	<0.0001
	Total	8	0.41 (0.29, 0.52)	<0.0001	98.0	<0.0001
	Mean age					
	<50 old	5	-0.52 (-0.73, -0.32)	<0.0001	93.8	<0.0001
	≥50 old	3	0.81 (0.68, 0.95)	<0.0001	98.8	<0.0001
	Treatment duration					
	<24 weeks	2	-1.11 (-1.51, -0.72)	<0.0001	59.0	0.118
Fat mass>	≥24 weeks	6	0.54 (0.42, 0.66)	<0.0001	98.2	<0.0001
	Total	12	-0.23 (-0.31, -0.25)	<0.0001	62.7	0.002
	Mean age					
	<50 old	7	-0.07 (-0.19, 0.06)	0.225	00.0	0.799
	≥50 old	5	-0.37 (-0.48, -0.26)	<0.0001	71.4	0.007
	Total	12	-0.12 (-0.21, -0.04)	0.006	95.3	<0.0001
	Mean age					
	<50 old	7	-0.26 (-0.41, -0.11)	0.001	97.3	<0.0001
	≥50 old	5	-0.05 (-0.16, 0.06)	0.413	48.7	0.099
	Total	12	-0.34 (-0.43, -0.24)	<0.0001	90.1	<0.0001
SBP	Mean age					
	<50 old	4	-0.06 (-0.28, 0.16)	0.601	96.9	<0.0001
	≥50 old	8	-0.40 (-0.51, -0.29)	<0.0001	6.9	0.377
	Treatment duration					
	<24 weeks	4	-0.48 (-0.81, -0.14)	0.006	22.1	0.276
	≥24 weeks	8	-0.32 (-0.42, -0.22)	<0.0001	93.4	<0.0001
	Type of disease					
	Obesity	8	-0.32 (-0.42, -0.22)	<0.0001	93.4	<0.0001
	Other	4	-0.44 (-0.77, -0.10)	0.0100	12.4	0.331
	Total	14	-0.78 (-0.87, -0.68)	<0.0001	95.2	<0.0001
DBP	Mean age					
	<50 old	5	-0.16 (-0.34, 0.02)	0.081	80.6	<0.0001
	≥50 old	9	-1.03 (-1.14, -0.91)	<0.0001	96.2	<0.0001
	Treatment duration					
	<24 weeks	4	-0.57 (-0.91, -0.23)	0.001	63.8	0.040
	≥24 weeks	10	-0.79 (0.90, -0.69)	<0.0001	96.6	<0.0001
	Type of disease					
	Obesity	10	-0.79 (-0.90, -0.69)	<0.0001	96.6	<0.0001
	Other	4	-0.57 (-0.91, -0.23)	0.001	63.8	0.040
	Total	9	-0.26 (-0.37, -0.16)	<0.0001	98.2	<0.0001
FBG	Mean age					
	<50 old	3	1.45 (1.09, 1.82)	<0.0001	99.3	<0.0001
	≥50 old	6	-0.42 (-0.53, -0.31)	<0.0001	91.0	<0.0001
	Treatment duration					
	<24 weeks					
	≥24 weeks					
	Type of disease					
	Obesity					
	Other					
	Total					
Insulin	Mean age					
	<50 old					
	≥50 old					
	Treatment duration					
	<24 weeks					
	≥24 weeks					
	Type of disease					
	Obesity					
	Other					
	Total					
HOMA-IR	Mean age					
	<50 old					
	≥50 old					
	Treatment duration					
	<24 weeks					
	≥24 weeks					
	Type of disease					
	Obesity					
	Other					
	Total					

(continued on next page)

Table 3 (continued)

Variable	Group	No. of trials	SMD (95 %CI)	P-value	I ² (%)	P-heterogeneity
TC	<24 weeks	2	-1.41 (-1.94, -0.87)	<0.0001	87.6	0.005
	≥24 weeks	7	-0.22 (-0.33, -0.11)	<0.0001	98.6	<0.0001
	Type of disease					
	Obesity	7	-0.22 (-0.33, -0.11)	<0.0001	98.6	<0.0001
	Other	2	-1.41 (-1.94, -0.87)	<0.0001	87.6	0.005
	Total	13	0.36 (0.26, 0.45)	<0.0001	93.7	<0.0001
	Mean age					
	<50 old	7	0.87 (0.68, 1.06)	<0.0001	95.7	<0.0001
	≥50 old	6	0.19 (0.08, 0.29)	0.001	63.9	0.0017
	Treatment duration					
LDL-C	<24 weeks	2	-0.77 (-1.25, -0.28)	0.002	00.0	0.563
	≥24 weeks	11	0.40 (0.31, 0.50)	<0.0001	94.1	<0.0001
	Type of disease					
	Obesity	9	0.41 (0.31, 0.50)	<0.0001	94.4	<0.0001
	Other	11	-0.77 (-1.25, -0.28)	0.002	00.0	0.563
	Total	13	0.21 (0.12, 0.31)	<0.0001	97.5	<0.0001
	Mean age					
	<50 old	7	0.71, (0.49, 0.92)	<0.0001	98.5	<0.0001
	≥50 old	6	0.09 (-0.002, 0.20)	0.116	90.2	<0.0001
	Treatment duration					
HDL-C	<24 weeks	2	-0.76 (-1.25, -0.28)	0.002	00.0	0.908
	≥24 weeks	11	0.25 (0.16, 0.36)	<0.0001	97.8	<0.0001
	Type of disease					
	Obesity	11	0.25 (0.16, 0.36)	<0.0001	97.8	<0.0001
	Other	2	-0.76 (-1.25, -0.28)	0.002	00.0	0.908
	Total	14	-0.33 (-0.41, -0.24)	<0.0001	94.5	<0.0001
	Mean age					
	<50 old	8	-0.01 (-0.14, 0.15)	0.981	92.7	<0.0001
	≥50 old	6	-0.54 (-0.66, -0.43)	<0.0001	95.3	<0.0001
	Treatment duration					
TG	<24 weeks	2	-0.47 (-0.94, 0.01)	0.052	00.0	0.519
	≥24 weeks	12	-0.32 (-0.41, -0.23)	<0.0001	95.4	<0.0001
	Type of disease					
	Obesity	12	-0.32 (-0.41, -0.23)	<0.0001	95.4	<0.0001
	Other	2	-0.47 (-0.94, 0.01)	0.052	00.0	0.519
	Total	11	-0.29 (-0.38, -0.19)	<0.0001	83.6	<0.0001
	Mean age					
	<50 old	9	-0.30 (-0.40, -0.19)	<0.0001	86.5	<0.0001
	≥50 old	2	-0.25 (-0.49, -0.02)	0.037	38.8	0.201
	Total	6	0.40 (0.29, 0.52)	<0.0001	97.7	<0.0001
Adiponectin	Mean age					
	<50 old	3	0.74 (0.49, 0.99)	<0.0001	98.9	<0.0001
	≥50 old	3	0.32 (0.19, 0.42)	<0.0001	90.9	<0.0001
Resistin	Total	6	1.08 (0.96, 1.20)	<0.0001	98.8	<0.0001
	Mean age					
	<50 old	3	-0.51 (-0.72, -0.30)	<0.0001	90.6	<0.0001
Leptin	≥50 old	3	1.90 (1.74, 2.05)	<0.0001	97.5	<0.0001
	Total	6	-0.57 (-0.68, -0.46)	<0.0001	92.6	<0.0001
	Mean age					
CRP	<50 old	3	-1.12 (-1.35, -0.89)	<0.0001	94.6	<0.0001
	≥50 old	3	-0.40 (-0.53, -0.27)	<0.0001	00.0	0.529
	Total	6	-0.38 (-0.49, -0.27)	<0.0001	84.1	<0.0001

BMI: body mass index, WC: waist circumference, WHtR: waist-to-height ratio, SBP: systolic blood pressure, DBP: diastolic blood pressure, FBG: fasting blood glucose, HOMA-IR: homeostatic Model Assessment for Insulin Resistance, TC: total cholesterol, LDL-C: low-density lipoprotein-cholesterol, HDL-C: high-density lipoprotein-cholesterol, TG: triglyceride, CRP: C-reactive protein, SMD: Standard mean difference, 95%CI: 95 % confidence interval.

weeks) influenced the heterogeneity of the findings. In people over 50 years of age, LCHPD had a greater effect on indicators such as weight loss, improved glucose metabolism, and reduced blood pressure, while in younger people, greater changes were observed in fat mass and inflammatory markers. Besides, shorter periods of the diet (less than 24 weeks) usually show weight loss and improvements in blood lipid indicators, but for a sustained reduction in insulin levels and WHtR, longer periods (more than 24 weeks) are more effective. These findings suggest that adjusting the diet based on individual characteristics such as age and duration can yield better results. In line with our subgroup analysis, a systematic review and meta-analysis examining long-term weight maintenance following LCHPD found that the impact of this dietary approach on metabolic indicators like weight and fat mass was reduced over 6 months [35]. A meta-analysis of long-term follow-up (>24 weeks) regarding high-protein diets also found inconsistent and uncertain weight loss effects [64]. This may be due to reduced adherence over

the long term and a lack of support to stay on track. Also, the long-term reduction in insulin levels is to weight loss, as it often occurs as a secondary effect of reduced body weight [65].

The metabolic factor-improving mechanisms behind LCHPD are multifaceted and interconnected. Reducing carbohydrate intake and increasing dietary protein helps reduce insulin spikes and improve insulin sensitivity, thereby improving blood sugar regulation [66,67]. Blood sugar regulation plays a key role in appetite control and preventing energy overconsumption [68]. Another appetite-regulating mechanism for LCHPD is the increased secretion of satiety hormones such as peptide YY (PYY), cholecystokinin (CCK), and Glucagon-like peptide-1 (GLP-1), which signal the brain to reduce hunger [69,70]. In addition, dietary protein has the highest thermogenesis compared to other macronutrients, which leads to an increase in calorie expenditure [71]. In an 8-week crossover study of 17 obese men, a low-carb, high-protein ketogenic diet reduced hunger, energy intake, and weight loss

significantly more than a moderate-carb non-ketogenic diet, likely due to increased satiety caused by elevated ketones in plasma and urine [72].

Our meta-analysis revealed that LCHPD increased TC and LDL-C while reducing HDL-C. Although triglyceride levels showed a favorable reduction. The rise in TC and LDL-C could be linked to the saturated fat found in animal-based sources, which accounts for a large proportion of the protein intake [73]. A similar effect was evident in a previous meta-analysis, in which a low-carbohydrate, low-fat diet with relatively high protein levels also increased TC and LDL-C, while HDL-C and TG levels showed considerable improvements [74]. The beneficial impact of a low-carb, high-protein diet on lowering triglycerides is partially due to the decrease in simple carbohydrates containing fructose. Regardless of insulin and phosphofructokinase modulation, fructose increases hepatic triglyceride synthesis and secretion [75].

Another meta-analysis that evaluated the effects of higher-protein diets compared with lower-protein diets on cardiometabolic risk factors also observed a reduction in TC and TG levels without a positive effect on LDL-C and HDL-C. In addition, this study, in line with our findings, observed a significant reduction in systolic blood pressure, weight, and fat mass. They found the reduction in blood pressure to be due to weight loss [76]. In a 26-week study of people who lost more than 8 % of their body weight over 8 weeks, blood pressure rebounded less in those on a high-protein diet than in those on a low-protein diet. This study also attributed the effect primarily to weight changes rather than solely to protein intake [77].

Studies targeting the effects of low-carbohydrate and/or high-protein diets on levels of energy-regulating hormones (e.g. leptin, adiponectin, and resistin) have not produced entirely consistent results. Aligned with our research, a study on individuals with varying GLP-1 genotypes (GG, GA, and AA) found that a long-term intervention with a low-carbohydrate, high-protein diet led to a reduction in metabolic markers such as leptin, TC, LDL-C, insulin levels, and HOMA-R, regardless of their genotype [50]. Also, in a systematic review and dose-response meta-analysis of 11 randomized controlled trials, low-carbohydrate diets (LCD) were shown to modestly but significantly increase adiponectin levels compared with control diets (0.02 µg/mL, 95 % CI: 0.01 to 0.03). Subgroup analysis showed that diets with <30 % of energy from carbohydrates significantly increased adiponectin, whereas those with ≥30 % carbohydrates had no effect [78]. Conversely, another systematic review and meta-analysis found that low-carb diets had no meaningful overall effect on leptin and adiponectin levels, although a slight increase in adiponectin was observed in younger subjects and with longer intervention duration [79]. The difference between our results and those of this meta-analysis may be due to less strict study selection criteria as well as the inclusion of any low-carbohydrate diet (regardless of whether it was combined with a high protein diet or ketogenic/paleo diets), as mentioned in the first-mentioned meta-analysis. The decrease in fat mass seems to be the primary factor leading to higher levels of adiponectin and lower levels of leptin and resistin [72,80]. Additionally, improvements in inflammatory markers and decreased insulin resistance may also play a role in these changes [81–83].

A low-carb diet, by lowering blood sugar and stimulating ketone production, reduces insulin secretion and fat storage, which ultimately reduces inflammation [84]. This reduction in blood sugar also reduces immune cell activation and the secretion of inflammatory cytokines such as tumor necrosis factor-α (TNF-α) and interleukin-6 (IL-6), helping to reduce CRP levels [85,86]. This CRP-lowering effect was also evident in our study. The anti-inflammatory effects of high-protein diets depend on the type of protein. Plant proteins have better anti-inflammatory effects as they help reduce oxidative stress and inflammatory cytokines [87]. Conversely, animal proteins, due to their high content of saturated fatty acids and methionine, may increase the production of reactive oxygen species (ROS), leading to an increase in CRP levels [88,89]. Santos et al. conducted a systematic review and meta-analysis on the impact of

low-carbohydrate diets on cardiovascular health. Their findings indicated significant decreases in CRP levels, as well as reductions in body weight, blood pressure, triglycerides, and fasting sugar [90]. Additionally, in a 3-month clinical trial comparing the effects of low-carbohydrate versus low-fat diets in 90 obese, non-diabetic adults, the low-carbohydrate diet was found to be more effective in reducing CRP levels, highlighting its anti-inflammatory effects compared to the low-fat diet [91]. On the other hand, during a 6-month intervention involving a low-calorie, low-fat diet for overweight individuals, a small reduction in CRP levels was observed in both groups. However, no significant difference was observed between the groups that followed the low-carbohydrate, high-protein diet compared to the high-carbohydrate, low-protein diet. The study concluded that the persistently high prevalence of overweight and obesity may have contributed to the lack of significant reduction in CRP levels [92].

5. Limitations

Our study may have several limitations that need to be addressed. Due to the dietary intervention design and the inevitable lack of blinding, selection bias may have affected the outcome. Also, the high heterogeneity between the included studies necessitates the examination of subgroups for the interpretation and generalizability of the results. Variations in the conditions of the study populations (such as patients with diabetes, PCOS, and gastric bypass surgery), age range, and length of intervention periods could have led to this heterogeneity. Additionally, the current meta-analysis considered only the CRP as a representative of general inflammation, overlooking other factors that may also reflect inflammation, such as TNF-α, IL-6, and interleukin-10 (IL-10). Another limitation was that, considering the effect of carbohydrate quality (complex or simple) and protein source (animal or vegetable) on metabolic indices, it would have been better to include these items in the subgroup analysis, although, due to the lack of reporting these items in the included studies, we were also unable to report them.

This study also had several strengths, as follows: Based on the available evidence, it is the first systematic review and meta-analysis to comprehensively examine clinical trials of low-carbohydrate, high-protein diets on metabolic and hormonal markers. In addition, conducting subgroup analyses to investigate the effects of mean age and duration of follow-up is another strength, which has increased the depth and validity of the results. Finally, the sensitivity analysis showed that none of the included articles had a significant impact on the combined effect size of the variables, and no publication bias was observed for these indices.

6. Conclusion

The findings of this systematic review and meta-analysis of 15 randomized clinical trials showed that LCHPD had a significant reducing effect on weight, BMI, WHtR, SBP, DBP, FBG, insulin, HOMA-IR, TG, and CRP. In addition, it increases adiponectin, but has no beneficial effect on other factors. Subgroup analysis showed that the effects of this diet were greater in people over 50 years of age and in short-term treatments. These results highlight the clinical importance of LCHPD as a potential short-term dietary strategy to improve metabolic health, particularly in older adults. Although, findings of the current systematic review and meta-analysis can't be generalized to entire societies, furthermore clinician should be care about the cultural and preferences of people. In addition, more studies are recommended to confirm these results and investigate long-term effects.

Authors' contributions

Conceptualization: O.N., M.N.

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Formal analysis: O.N.

Funding acquisition: Not applicable.
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Funding

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors confirmed that no third-party contributions were involved in the present research.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dsx.2025.103352>.

Data availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

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